METRIC

MIL-S-24725/2
29 November 1994

MILITARY SPECIFICATION SHEET

SWITCH, FIBER OPTIC, NONHARD MOUNTED, ELECTRICAL NONLATCHING; BYPASS, MULTIMODE CABLE, STAND-ALONE (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24725.

CLASSIFICATION

Type: 4 (Electrical nonlatching).
Style: C (2:2 bypass).

DESIGN AND CONSTRUCTION

Electrical pigtails: Applicable.

Fiber type: In accordance with MIL-F-49291/6.

Optical pigtail: In accordance with MIL-C-85045/14 or MIl-C-85045/16.

Actuation method: Electrical nonlatching.

Pin terminals: Not applicable.

Mounting dimensions: See figure 1.

Seals: Not applicable.

Enclosures: Watertight.

Dimensions: See figures 2 and 3.

Weight: ≤ 180 grams (does not include mounting hardware).

Materials: Magnetic materials may be used internal to the switch actuator construction.

Color: Gray.

Finish: Conversion coat in accordance with MIL-C-5541, class 3, conductive.

Paint: Paint all surfaces except base.

Marking: Either the switch case, or each pigtail shall be marked to indicate: Node IN, Node OUT, RING IN, RING OUT, +V, and Return. If each pigtail is marked, it shall be marked on heat shrink tubing applied to the pigtail as it exits the switch.

Mounting hardware: Supplied with the switch, 4M1 by 12 mm round head machine screws.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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PERFORMANCE REQUIREMENTS

Optical properties:

Insertion loss: See figure 4: Node output to node input: plus 1 to plus 14 decibels (dB).
Other paths: ≤ 0.8 dB.

Operating time plus operating bounce time: ≤ 25 ms.

Release time plus release bounce time: ≤ 25 ms.

Operating life switching rate: ≤ 20 Hz.

Mechanical properties:

Connectorized: Not applicable.

Environmental properties:

Temperature range: Operating: -28°C to +65°C
Storage: -40°C to +70°C

Hermetic: Not applicable.

Sand and dust: Not applicable.

Electromagnetic effects: Applicable, below deck applications.

Electrical properties:

Electrical input signals:

Voltage and current levels: 4.75 to 5.5 volts, ≤ 240 milliamperes (mA) maximum.

Shock: Applicable; sample units shall be mounted on a secondary plate which is shock isolation mounted to the 4A plate. The secondary plate mass shall be not greater than 1.45 kg. The shock isolation shall be achieved using four Aeroflex (C2-H-510) shock isolators or equivalent.

Nuclear radiation resistance: Nuclear radiation resistance test conditions shall be as shown below and in table I.

Light launch conditions: In accordance with EIA-455-50.
Wavelength: 1300 ±25 nm.
Source type: LED with FWHM spectral width ≤ 160 nm.

TABLE I. Steady-state gamma radiation test condition.

Test temperature (°C)	Dose rate	Total dose [rad (Si)]
-28 ±2 25 ±2 65 ±2	50 +0, -20 rad (Si)/sec	Classified

(Some nuclear radiation resistance characteristics of this optical fiber are classified and shall be obtained from the qualifying activity. Application to receive these requirements must be made through the US Naval Sea Systems Command, Attn: SEA 03K31, Washington, DC 20362. Information concerning security clearance classification and "need to know" must be detailed in the request.)

QUALITY ASSURANCE PROVISIONS

Switch position: For mechanical and environmental tests switches shall be tested in the activated position with voltage applied and in the bypass position. Each switch shall be tested in 1 position in each test.

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Electrical properties inspection:

Voltage and current levels:

Step 1: The minimum voltage level shall be applied to the switch. Observe that the switch changes state and verify the insertion loss for both channels. Measure and verify the current.

Step 2: Reduce the voltage to zero volts. Verify that the switch changes state. Verify the insertion loss for both channels.

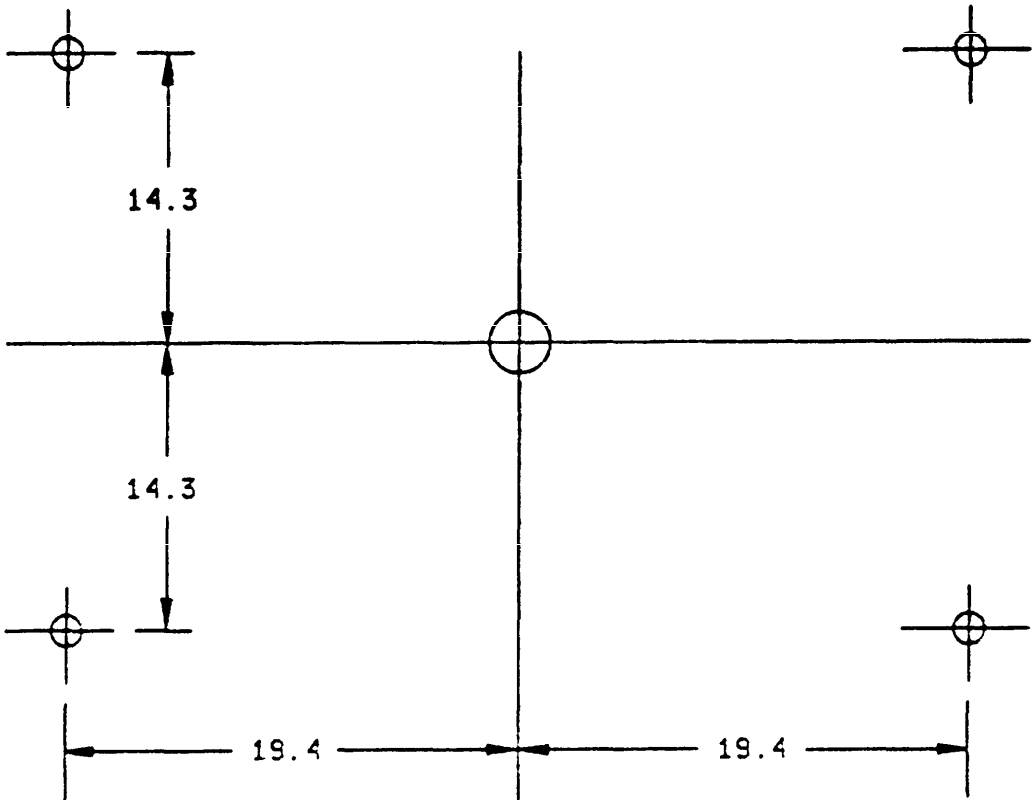
Step 3: Repeat steps 1 and 2 for the nominal voltage level (5.125 volts).

Step 4: Repeat steps 1 and 2 for the maximum voltage (5.5 volts).

Part or Identifying Number (PIN): M24725/2-01T (Switches with optical pigtails entering on opposing sides).
M24725/2-01S (Switches with optical pigtails entering on 1 side).

Intended use: These switches are intended for use in shock isolated configurations inside protective enclosures

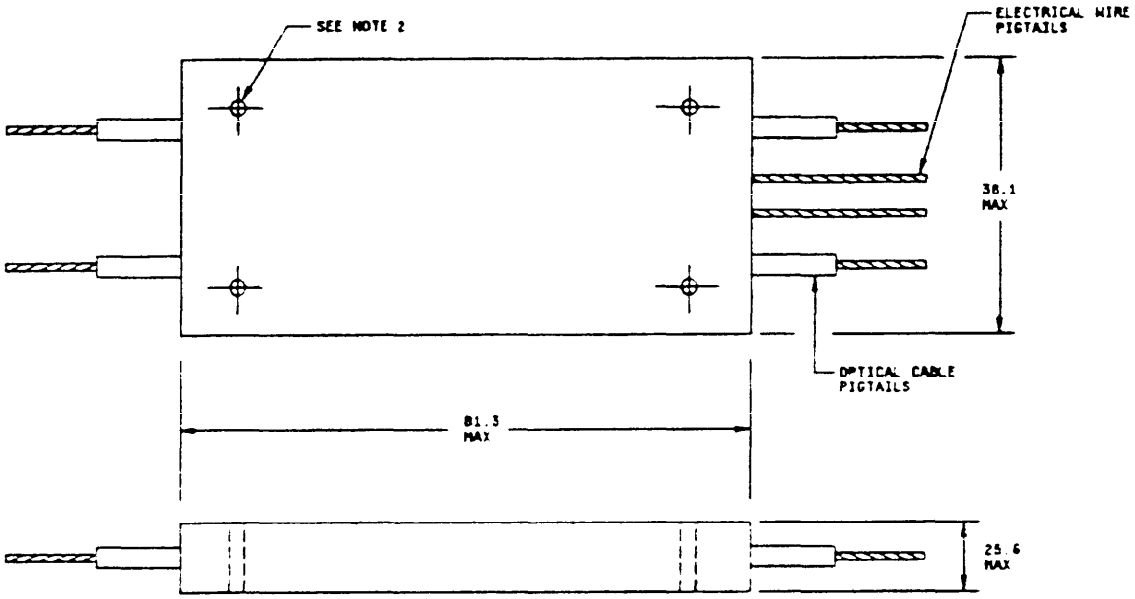
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NOTES:
1. Dimensions are in millimeters.
2. Unless otherwise specified, tolerance is 0.3 mm.

FIGURE 1. Mounting dimensions.

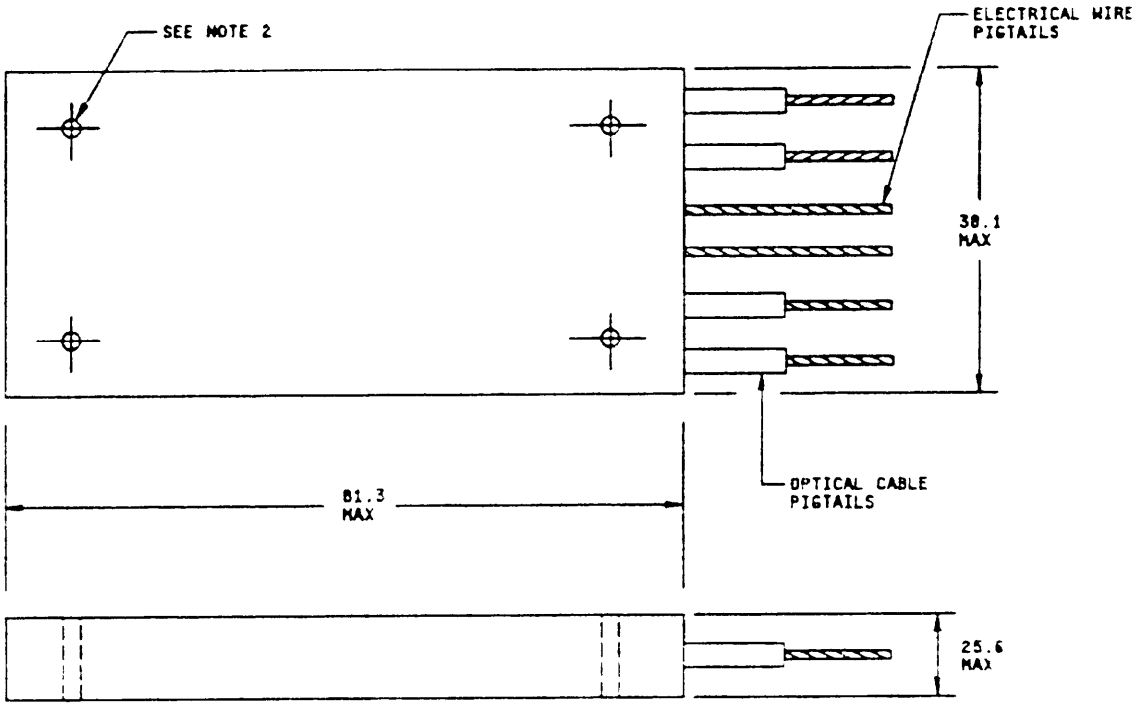
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NOTES:
1. Dimensions are in millimeters.
2. M1 inserts (alternate footprint to have countersunk through holes sufficient to pass M1 machine screws)(see note 3).
3. Switch must mate with mounting dimensions shown on figure 1. Switches are not required to have all mounting holes.
4. Pigtail locations are for reference only.

FIGURE 2. 2 x 2 Bypass switch, M24725/2-01T.

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NOTES:
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2. M1 inserts (alternate footprint to have countersunk through holes sufficient to pass M1 machine screws)(see note 3).
3. Switch must mate with mounting dimensions shown on figure 1. Switches are not required to have all mounting holes.
4. Pigtail locations are for reference only.

FIGURE 3. 2 x 2 Bypass switch, M24725/2-01S.

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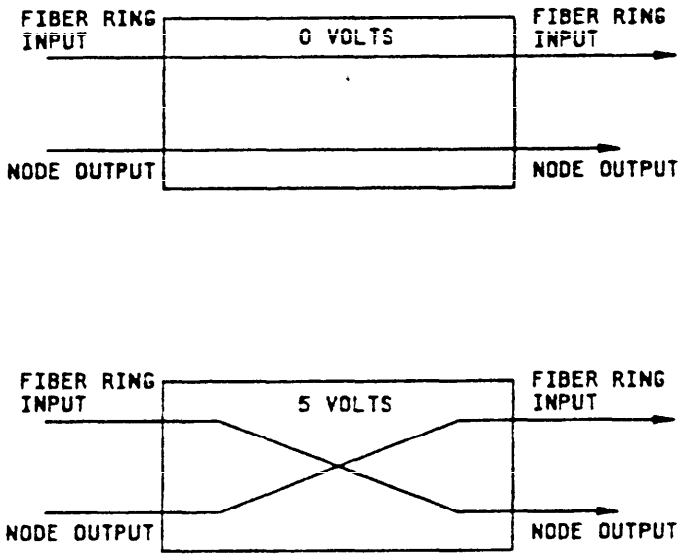


FIGURE 4. 2 x 2 Bypass switch - functional diagram.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - MI, SC
Navy - AS
Air Force - 17
DLA - ES

Preparing activity:
NAVY - SH

Agent:
DLA - ES

(Project number: 6021-0002)

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